Canto · one-time blind test

Displaced-track reconstruction, trained on ttbar only · frozen models, evaluated once on the test sets

	Canto-L 3 seeds · TEST	Canto-H 3 seeds · TEST	Canto-B 3 seeds · TEST	acorn (HNL) 3 seeds · TEST	acorn 3 seeds · TEST
Displaced tracks |d0| ≥ 10 mm or r_v ≥ 100 mm	99.1%	**99.2%**	98.4%	94.5%	90.4%
Far-displaced tracks |d0| 100–300 mm	**98.6%**	98.4%	97.0%	93.1%	84.9%
Very far-displaced |d0| ≥ 300 mm	87.0% 94/108 particles	**91.7%** 99/108 particles	89.8% 97/108 particles	**91.7%** 99/108 particles	79.6% 86/108 particles
Prompt muons W → μ in HNL events	**99.8%**	**99.8%**	**99.8%**	98.5%	98.4%
Dense prompt tracks ttbar, |d0| < 1 mm	96.3%	96.2%	**96.6%**	84.5%	87.4%
Fake-track rate lower is better	**1.02%**	1.42%	1.29%	2.90%	2.95%
Training data	ttbar only	ttbar only	ttbar only	HNL signal	ttbar only
Parameters edge classifier	348k	348k	348k	264k	264k

All numbers: ACTS generic detector simulation without pileup, the one-time blind test of the frozen models: the displaced-track, prompt-muon and fake-rate rows from the pooled HNL test sets (hnl10_100_test, hnl15_200_test and hnl15_100_external, 6,999 events), 'Dense prompt tracks' from ttbar_test (1,000 events); track efficiency by double-majority matching; mean over the 3 training seeds, as in the preregistered claims. |d0| 100 –300 mm: 584 and |d0| ≥ 300 mm: 36 HNL daughters per seed, the latter statistics limited. Best value per row bold and shaded (ties included).

Frozen before the test: the Canto variants, every seed's builder configuration and the working points were committed before any Canto test job (results/final_selection_v1.json, commit aecb23f), and every acorn cut was fixed on val; nothing was chosen on test. Canto (trained on ttbar only, full budget; the branches differ in the graph: Canto-B the base graph, Canto-H a hand-cut helix-triplet union, Canto-L a learned triplet-support union) at the builder configurations chosen on val by rule R6 with the one-standard-error rule: Canto-L learned merging, Canto-H walkthrough + merging, Canto-B learned fragment merging. The highlighted Canto-L is the variant selected on val (rule R8); it is not re-chosen on test. These are post-freeze summaries of the one-time evaluation; the preregistered claims (results/claims_test_v1.json) are unchanged.

acorn (HNL): acorn's interaction network with connected components (CC), trained on the HNL signal as in Wang et al. (arXiv:2203.08800), whose own pipeline used an attention GNN and DBSCAN; it reproduces their all-track efficiency (test: 0.9554 vs 0.9561 in the paper). acorn: the same network trained on ttbar only, like Canto. Both at the cut chosen as best on val (the highest efficiency at a fake rate ≤ FR* = 2.94 % on pooled HNL val, else the lowest-fake-rate cut), applied unchanged to test: acorn (HNL) 0.38 / 0.36 / 0.44, acorn 0.34 / 0.33 / 0.31 (seeds 0 / 1 / 2; acorn (HNL) seeds 0 and 2: CC at these cuts on the stored test scores after the freeze, as the one-time test chain did not include them). With the preregistered cut rule, the reference of the claims (acorn (HNL) 0.26 / 0.36 / 0.27; acorn the same cuts, so the values shown): acorn (HNL) 92.6 % displaced, 91.7 % at |d0| 100 –300 mm, 98/108 at ≥ 300 mm, 97.6 % prompt muons, 81.7 % dense prompt tracks, fake rate 2.86 %.

DeZoort IN (and IN-L) was never evaluated on test and has no column. FR* is the working point on val; the test fake rates are as measured.